

Within-host mathematical models of hepatitis B virus infection: Past, present, and future

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Abstract

Mathematical modeling has been instrumental in enhancing our understanding of the viral dynamics of hepatitis B virus (HBV) infection. We give a primer on HBV infection in humans and a brief overview of the development of within-host mathematical models of HBV infection. In the last decade, models have advanced from considering chronic HBV infections under therapy to the pathogenesis of infection. We also summarize estimates of key viral dynamic parameters that have varied greatly among studies and show that they impact model predictions. Future directions for mathematical modeling of HBV infection are proposed to better understand emerging therapies, the HBV life cycle, prediction of cure, and the mechanisms involved in the immune response to HBV infection.

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Hepatitis B virus, Mathematical modeling, Viral dynamics, Cure, Covalently closed circular DNA.

Introduction

Hepatitis B virus (HBV) infection is an underfunded and overlooked disease. Despite the availability of an effective vaccine for the past three decades, there are still more than 240 million chronic carriers and ~1 million disease-related deaths recorded worldwide every year [1]. There is currently no cure for HBV infection, but novel therapies are being investigated [2].

Mathematical modeling has served as a powerful tool to study viral dynamics and immune responses and to

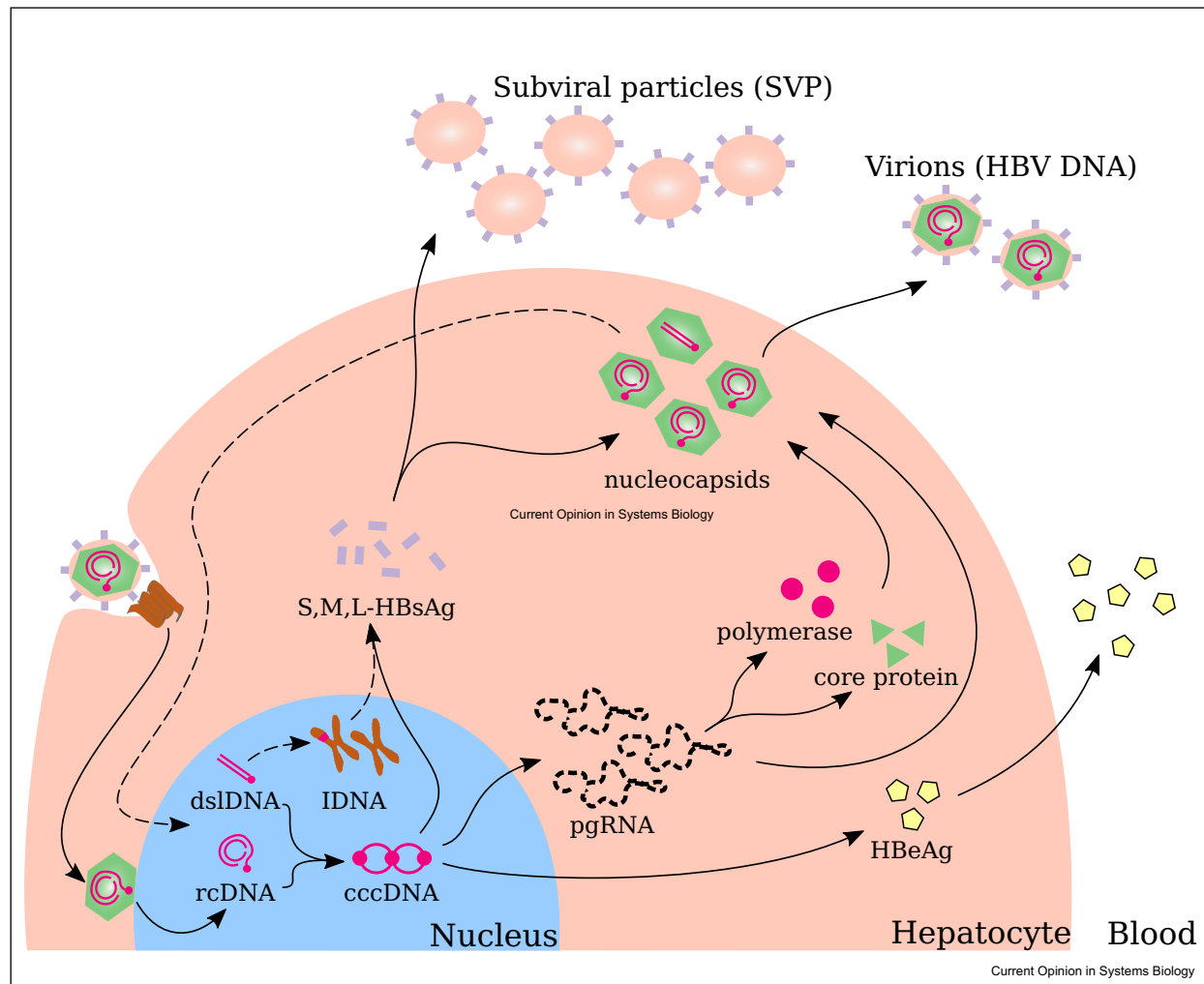
evaluate treatment strategies [3–5]. This review gives a brief overview of mathematical models of the viral dynamics of HBV infection in vivo, followed by an in-depth discussion of quantitative estimates of key viral infection parameters and their importance to mechanistic models of HBV infection and to predictions of the success or required duration of therapy.

The life cycle of HBV

HBV is a noncytopathic enveloped virus that is ~42 nm in diameter. Each virion consists of a nucleocapsid surrounded by a lipid bilayer that is studded with three viral glycoproteins, called small, medium, and large hepatitis B surface antigens (S-,M-,L-HBsAg) [6]. Within the nucleocapsid is a circular, partially double-stranded DNA, with an incomplete positive strand, called relaxed circular DNA (rcDNA), complexed with a DNA polymerase [7].

HBV infects hepatocytes, the parenchymal cells of the liver [6]. The virus life cycle starts when a virion attaches to a cell surface receptor, followed by entry and release of the nucleocapsid into the cytoplasm (Figure 1). The nucleocapsid is transported to a nuclear pore, where it degrades and releases rcDNA into the nucleus [6,7]. Within the nucleus, rcDNA is repaired by host enzymes and is converted into covalently closed circular DNA (cccDNA) [8]. The cccDNA serves as a template for the transcription of pregenomic RNA (pgRNA) and precore and subgenomic mRNAs, whose translation yields HBV polymerase and core proteins, hepatitis B e-antigen (HBeAg), and S-,M-,L-HBsAg, respectively. Immature nucleocapsids are assembled from core protein dimers that enclose a pgRNA polymerase complex and become mature nucleocapsids once pgRNA is reverse transcribed into DNA. After reverse transcription, 90% of nucleocapsids contain rcDNA, whereas the remaining ~10% contain double-stranded linear DNA (dslDNA) [9]. At this stage, nucleocapsids can either gain an envelope with HBsAg by passing through the postendoplasmic reticulum, pre-Golgi compartment and be secreted as virions into the blood or recycle back to the nucleus [7,8]. Virions are quantified as the amount of HBV DNA per milliliter in serum/plasma. Recycled dslDNA can integrate into the host genome becoming integrated DNA (IDNA), while recycled rcDNA and

Figure 1



HBV life cycle. HBV infects hepatocytes by initiating entry via the sodium taurocholate cotransporting polypeptide (NTCP) receptor and releasing its nucleocapsid into the cytoplasm. After nuclear import of relaxed circular DNA (rcDNA), rcDNA is converted into covalently closed circular DNA (cccDNA), which acts as a template for the transcription of pregenomic RNA (pgRNA), precore mRNA, and other subgenomic mRNAs. In the cytoplasm, translation of pgRNA and mRNAs produces HBV polymerase, core protein, hepatitis B e-antigen (HBeAg), and S,M,L-HBsAg. Immature nucleocapsids are assembled from core proteins that enclose a pgRNA–polymerase complex. As a consequence of reverse transcription of pgRNA, 90% of nucleocapsids contain rcDNA, whereas the remaining ~ 10% contain double-stranded linear DNA (dslDNA). Nucleocapsids can either become enveloped by passing through the postendoplasmic reticulum, pre-Golgi compartment and be secreted as virions into the blood (referred to as **HBV DNA**) or be recycled back to the nucleus (dashed arrow). Recycled dslDNA can integrate into the host genome (which we refer to as integrated DNA [IDNA]), while both recycled rcDNA and dslDNA can amplify the number of cccDNA in the nucleus. Concurrently, subviral particles (**SVPs**), which are similar to virions that are devoid of rcDNA/dslDNA, are secreted in excess via multivesicular bodies. Even in the absence of cccDNA, an infected cell with IDNA can produce SVPs. Finally, infected cells also secrete HBeAg into the blood, which is believed to induce immune tolerance. HBV, hepatitis B virus.

dslDNA can further amplify the number of cccDNAs in the nucleus [6,8].

An infected cell also secretes subviral particles (SVPs), also called HBsAg particles. SVPs are enveloped particles with S-,M-,L-HBsAg on their surface, but lack rcDNA/dslDNA [10]. Compared with virions, SVPs are secreted in great excess and may act as immune system decoys [6,10]. Even in the absence of cccDNA, an infected cell with IDNA can produce SVPs [9]. Finally,

HBeAg is also secreted by infected cells and is believed to induce immune tolerance [6,11].

Acute and chronic phases of HBV infection

When individuals become infected with HBV during adulthood, they go through an acute infection phase lasting <6 months. In the initial phase, HBV DNA levels are low, and there is poor induction of innate immune responses [12,13]. At the peak of acute infection, ~8–12 weeks after infection, high HBV DNA levels

($>10^9$ – 10^{10} copies/mL) and SVPs (at least 100-fold more plentiful than HBV DNA [6]) can be detected in serum, and almost all hepatocytes ($\sim 10^{10}$ – 10^{11} cells) in the liver are infected [6,12]. Concomitantly, there is an induction of adaptive immune responses involving CD4+ and CD8+ T cells and B cells. Cytokines released into the circulation can inhibit viral production, whereas the neutralizing activity of anti-envelope antibodies can reduce viral infectivity [12].

The immune response can also lead to the elimination of infected cells to clear the infection. When there is rapid elimination of infected cells, as in cases of acute injury, hepatocytes undergo multiple rounds of proliferation to restore liver mass [14]. When infected cells rapidly proliferate, it has been shown in vitro [15] and in vivo [16,17] that daughter cells lose cccDNA, such that there is a dilution of cccDNA in subsequent generations of cells. It is hypothesized that rapid proliferation does not allow rcDNA enough time to move from the cytoplasm to the nucleus to establish cccDNA in the daughter cells [15]. Thus, the clearance of acute HBV infection by the immune system is likely assisted by the loss of cccDNA during cellular proliferation, a possibility supported by a modeling study [18]. Overall, acute HBV infections are cleared in $>90\%$ of cases [12].

Acute infections that do not clear will transition into a chronic phase [12]. During chronic HBV (CHB) infection, HBV DNA levels are usually lower than those at the peak of acute infection [12]. HBV-specific T-cell responses also diminish [12], such that the elimination of infected cells does not occur as rapidly as in the acute phase. Thus, the loss of cells in CHB can be compensated by cellular proliferation at a rate slower than the rate during the clearance of acute HBV infection. An in vitro study showed that cccDNA does not become diluted in subsequent generations of nonexponentially proliferating cells [15]. Thus, when infected cells divide in CHB, cccDNA is likely passed to daughter cells, maintaining infection. CHB can persist for many years or even decades and may lead to liver cirrhosis, hepatocellular carcinoma, and death [6].

Treatment and cure of CHB infection

Treatment is mainly recommended for CHB because most acute infections resolve on their own [19]. The two currently approved therapies are nucleos(t)ide analogs (NAs), which act as reverse transcriptase inhibitors blocking rcDNA formation, and type I interferon, which inhibits HBV transcription and/or causes partial degradation of cccDNA [19–21]. Key indicators of the response to therapy are declines in serum concentrations of HBV DNA and HBsAg. Of particular note is that the serum HBsAg concentration represents the overall concentration of surface envelope proteins that are present on both serum HBV DNA and SVPs [22], and thus, this term

should not be confused with the concentration of serum SVPs. With either therapy, a typical outcome is HBV DNA being driven below assay detection limits, but if treatment is stopped, HBV DNA almost always rebounds [19,20]. At the same time, the level of serum HBsAg seldom decreases, indirectly reflecting the negligible impact of these therapies on cccDNA and/or IDNA, thus their failure to clear CHB [20]. Functional cure is defined as achieving sustained, undetectable HBV DNA and HBsAg in serum, after the cessation of treatment [20,23]. In a handful of cases, functional cure has been achieved in $<3\%$ and $\sim 10\%$ of patients under NA and type I interferon therapy, respectively [19,20]. The next therapeutic goals are to obtain a higher rate of functional cure and the complete eradication of intrahepatic cccDNA and IDNA to obtain a ‘complete sterilizing cure’ [23].

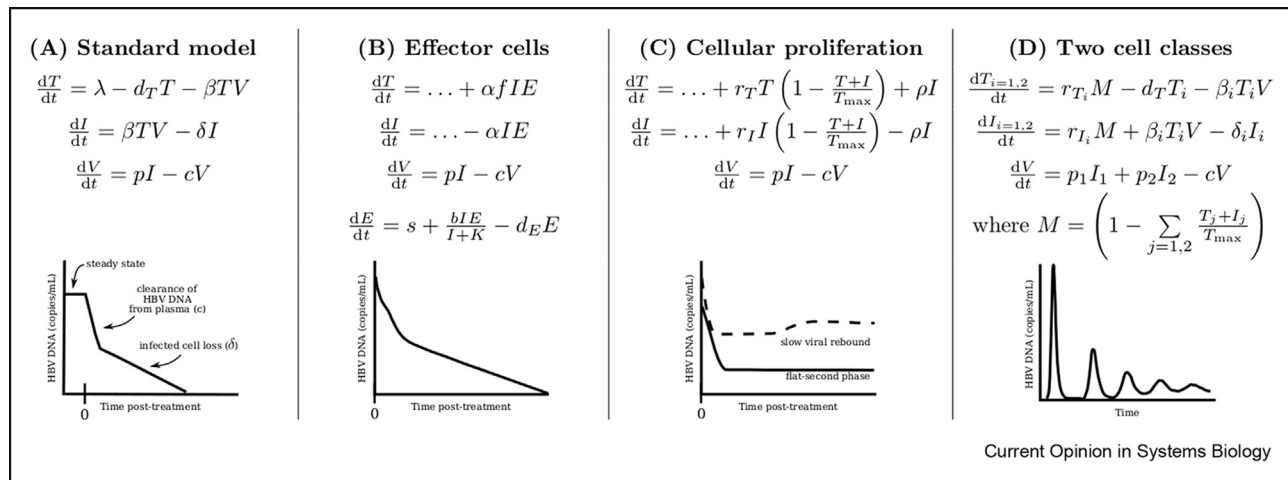
Development of mathematical models of HBV infection

Mathematical models allow us to quantify parameters of interest, test hypotheses, and estimate the efficacy or needed duration of a therapy. Over the past thirty years, progress in developing mathematical models of within-host HBV infection has been slow but steady (Figure 2).

After the success of simple viral dynamic models developed for analyzing HIV infection under therapy [24], a similar approach was taken for CHB [25–27] and for the clearance of acute infections [28]. The standard model is typically a system of ordinary differential equations with three compartments representing the concentrations of uninfected cells (T), productively infected cells or infected cells containing cccDNA and producing HBV DNA (I), and serum HBV DNA (V) (Figure 2a). When modeling CHB, the system is assumed to be at steady state before treatment, but this might not always be the case [29]. When NA treatment is modeled, this simple mathematical model can capture the biphasic decline observed in serum HBV DNA during therapy, where the slope of the first and second phase reflects the clearance rate of serum HBV DNA (ϵ) and the rate of productively infected cell loss (δ), respectively, if the effectiveness of therapy in blocking viral production is $\epsilon \sim 1$ (where $0 \leq \epsilon \leq 1$ and $\epsilon \sim 1$ corresponds to 100% effective therapy) [3,5]. From this analysis, quantitative estimates of the rates of processes in the HBV life cycle were obtained. For example, the half-lives of serum HBV DNA and productively infected cells were estimated as ~ 1 day and ~ 10 – 100 days, respectively [25–28]. The HBV DNA production rate was also estimated to be at least 10^{13} virions per day during acute HBV infection [28] and at least 10^{11} virions per day during CHB [27].

Additional mathematical models have been formulated to capture more complex patterns of serum HBV DNA decay during therapy [30–33]. A mathematical model that explicitly included effector cell dynamics was able to

Figure 2



Mathematical models of chronic HBV infection. (a) In the standard model of HBV infection [25], uninfected hepatocytes (target cells, T) are infected by HBV DNA (V) at a rate β . Target cells are supplied at a rate λ and die at a rate d_T . Productively infected cells (I) produce HBV DNA at a rate p per cell, until they die at a rate δ owing to immune responses. To model the treatment of chronic HBV infection, the virus is assumed to be at steady state (before treatment), and the effect of antivirals such as nucleos(t)ide analogs and interferon is modeled as a reduction in the viral production rate ($p = (1 - \varepsilon)\bar{p}$) or infection rate ($\beta = (1 - \eta)\bar{\beta}$), where η and ε represent the antiviral efficacy. On treatment, the standard model captures the biphasic decay of HBV DNA, where the slope of the first phase reflects the clearance rate of serum HBV DNA (c), and the slope of the second phase reflects the rate of productively infected cell loss (δ), if the effectiveness of therapy in blocking viral production is $\sim 100\%$ ($\varepsilon \sim 1$). (b) Variants of the standard model have added immune responses such as antibodies or effector cells [31,35,40]. In this example, effector cells E are supplied at a rate s , die at a rate d_E , and proliferate in a density-dependent manner with a maximum rate b , where the per-cell rate of proliferation is $(bI)/(I+K)$. Effector cells mediate infection by eliminating productively infected cells at a rate α or 'curing' a fraction (f) via noncytolytic processes. Accounting for effector cells during treatment of CHB allows the mathematical model to capture the triphasic decay of HBV DNA. (c) Other variants of the standard model have included the target and productively infected cell proliferation at rates r_T and r_I , respectively, where proliferation slows down as the total hepatocyte number ($T+I$) reaches its normal levels, $T_{\max}$ [18,30]. Curing of productively infected cells by noncytolytic processes has also been included in this mathematical model at a rate ρ . The mathematical model with proliferation terms was able to capture complex HBV DNA decay profiles such as slow viral rebound or a flat second phase observed during therapy, in addition to biphasic and triphasic decays. (d) Finally, a mathematical model with two uninfected and two productively infected cell classes (with contrasting responses to the virus [34]) can describe many features of untreated HBV infection over several years/decades, such as oscillations in HBV DNA during a phase of CHB called 'chronic active hepatitis.' Multiple cell classes may be motivated by various hepatocyte subtypes or heterogeneity in cccDNA levels in infected hepatocytes that cause differences in viral production rates or death rates. CHB, chronic HBV; HBV, hepatitis B virus.

describe the triphasic decay of serum HBV DNA during therapy [31] (Figure 2b). The inclusion of uninfected and infected cell proliferation in a mathematical model was also able to describe other patterns (e.g. stepwise decay, slow viral rebound, a flat second phase) in serum HBV DNA during therapy, in addition to triphasic decay [30] (Figure 2c). Other complex patterns seen in untreated CHB, such as oscillations in serum HBV DNA, could be captured in a mathematical model with two uninfected and two productively infected cell classes [34] (Figure 2d). Furthermore, oscillations in serum HBV DNA observed during therapy could also be explained using a mathematical model with expansion and contraction of multiple viral clones [33].

Mathematical models have also been used to further understand the pathogenesis of HBV infection [11,18,34–40]. Mathematical models have incorporated antibody responses [35,38], explored the interplay between two different viral clones to explain immune tolerance after prenatal HBV infection [11], and investigated the possibility of cell-to-cell spread of HBV infection [36]. Importantly, mathematical models

including intracellular HBV replication [36,37,39] and cellular proliferation during acute HBV infection [18] have also been proposed.

The variability in the estimated half-lives of serum HBV DNA and HBsAg

One component of complete cure is the elimination of all serum HBV DNA. Even one virion can potentially re-establish infection, causing viral rebound and treatment failure. During therapy, HBV DNA will fall below assay detection limits before it is known whether complete elimination occurs (i.e. whether HBV DNA falls below a threshold of <1 copy in the 15 L of extracellular fluid in the body [11]). In such cases, determination of the predicted time to achieve complete elimination is needed. Mathematical models rely on an accurate estimate of the half-life of HBV DNA to predict the needed duration of therapy.

Estimates of the half-life of HBV DNA from studies of CHB under antiviral therapy and acute infection range from several hours to several days (1–96 h) [25,26,28,32,41–45], with a mean of ~ 1 d (Table 1).

Table 1**Estimates of the serum half-life of HBV DNA.**

Estimate	Experimental condition	Reference
4.4 h	ADV	[43]
26.4 ± 7.2 h	ADV	[26]
9–24 h	LMV	[46]
26.4 h	LMV	[25]
55.2 ± 33.6 h	LMV/LMV+FCV	[47]
1–92 h	LMV/LMV+FCV	[32]
12–29 h	ETV	[45]
24–74 h	CLV	[31]
16.6 ± 1.9 h	IFN/IFN+LMV/LMV PEG-IFN/ PEG-IFN+LMV/ADV+LMV	[44]
3.4–30 h	PEG-IFN/PEG-IFN+LMV/LMV	[42]
2.4–69.2 h	PEG-IFN/PEG-IFN+LMV/LMV	[48]
28.8 ± 14.4 h	Acute infection ^a	[28]
38.4–96 h	Acute infection ^a	[41]

CHB, chronic HBV; HBV, hepatitis B virus.

The half-life of HBV DNA was estimated in studies of HBV infection in humans. Various therapies were given to patients with CHB, including adefovir (ADV), lamivudine (LMV), famciclovir (FCV), entecavir (ETV), clevudine (CLV), interferon (IFN), and pegylated interferon (PEG-IFN). The treatment groups within each study are denoted by a slash. Estimates from studies of acute infection are also shown. Estimates are expressed as a range (minimum–maximum), mean ± standard deviation, or mean.

^a A potential drawback in this study is that HBV DNA production could be continuing, and thus, this half-life estimate could be an overestimate.

However, studies that included an explicit description of intracellular replication suggest that these are overestimates of the half-life of HBV DNA [37,43].

Another component of complete or functional cure is the elimination of all circulating HBsAg in the serum. There is also high variability in the estimates of the half-life of serum HBsAg, ranging from a few hours up to 38 d [38,41,49–51], with a median of ~5 d (Table 2).

In this simplified example of one component of complete cure (i.e. elimination of HBV DNA), we made strong and unrealistic assumptions about the efficacy of antiviral therapy, illustrating the impact of these estimates on our predictions in an ideal case. Furthermore, we did not consider the elimination of productively infected cells (containing cccDNA), infected cells that only produce SVPs (infected cells containing only IDNA), HBsAg, or SVPs. The dynamics of infected cells and SVPs must also be considered when modeling functional or complete cure. The lack of confidence in the estimated half-life of HBV DNA and HBsAg impacts the ability to predict the needed duration of therapy. For example, with the standard model (Figure 2a), under the assumption that therapy completely blocks the production of HBV DNA, serum HBV DNA decays exponentially. The estimates of the half-life of HBV DNA of 1 h and 96 h result in vastly different predictions of the needed duration of therapy of 1.7 d and 163 d, respectively.

The variability in the estimated half-lives of productively infected cells and cccDNA

Elimination of cccDNA and IDNA is the basis of complete cure. To accurately predict the time to clearance of infection, the number of cccDNA copies per infected cell and the number of infected cells must be quantified, as well as their respective half-lives. Estimates of these quantities are also highly variable. Human liver biopsies have provided estimates of intrahepatic cccDNA between 0.02 and 14 copies per cell [52–55] and also as high as 147 copies per cell [56]. The half-life of cccDNA in humans with CHB has not been estimated [21], but one study suggests that cccDNA turnover is on a timescale of months [57]. In HBV-infected chimpanzees, it has been estimated as 9–14 d [58]. The fraction of productively infected hepatocytes in CHB has been estimated as ≤75% [59–61].

Furthermore, the percentage of hepatocytes with IDNA was estimated as 0.01–200 % in CHB [62]. The

Table 2**Estimates of the half-life of circulating HBsAg.**

Estimate	Experimental system	Drawback	Reference
5.3 ± 3.2 d	NAP	Only the abstract and poster is available.	[49]
5.7–8 d	Acute infection	HBsAg production could be continuing, and so this could be an overestimate.	[41]
38 d	LMV	Observed in only 3 of 10 patients. It could also be an overestimate as all production might not be blocked.	[38]
6.7 d	Heavy water labeling	The method of calculating the half-life of SVPs is based on the individual half-lives of S-, M-, and L-HBsAg.	[50]
0.2 d	HBV–HDV–coinfected patients receiving HBIG after transplantation	Identifiability issue with respect to separately estimating natural clearance from immune clearance	[51]

HBIG, hepatitis B immune globulin; HBV, hepatitis B virus; HDV, hepatitis delta virus; HBsAg, hepatitis B surface antigen; SVPs, subviral particles.

The half-life of serum HBsAg was estimated in studies of HBV infections in humans. In some studies, antiviral therapies were given, including lamivudine (LMV) and nucleic acid polymer (NAP) REP2139 monotherapy. Estimates are expressed as a range (minimum–maximum), mean ± standard deviation, or mean.

estimates of the half-life of productively infected cells in humans under antiviral therapy are also variable, ranging from 2.4 to 100 d [25,26,28,32,44–46], with a mean of 18 d [26] (Table 3). Again, the variability in these estimates impacts the predicted needed duration of therapy.

Future mathematical models

Although the desired goal of a therapy is to achieve functional or complete cure, most mathematical models have not accounted for serum HBsAg or SVPs along with intracellular cccDNA and IDNA [25,26,28,30,32,44–46]. Future models should explicitly consider these quantities and should be mindful of infected cell proliferation and immune responses in CHB.

Mathematical models that incorporate intracellular processes of the HBV life cycle

Additional insights into within-host HBV infection can be generated using multiscale mathematical models that incorporate intracellular processes. A few theoretical studies have explicitly modeled cccDNA [36,37,39] or IDNA [39], but only one of these studies has incorporated experimental data [37]. Multiscale models can be used to understand the mode(s) of action of new antivirals and their effectiveness in inhibiting various steps in the viral life cycle [63].

In developing such mathematical models for HBV infection, a key challenge is the collection of

quantitative intracellular data from humans (e.g. number of copies of pgRNA, rcDNA, and cccDNA). Liver biopsy is invasive, which means that sampling is infrequent and longitudinal data will be rarely available. Furthermore, a biopsy only gives a narrow view of infection in the whole liver. Depending on the availability of longitudinal intracellular data, mathematical modeling approaches similar to those in the studies by Goyal and Murray [36], Murray et al [37], Goyal and Chauhan [39], and Guedj et al [63] can be used.

Mathematical models that include infected cell proliferation

As discussed earlier, the proliferation of infected cells in CHB can maintain the infection and lengthen the time it takes to eradicate cccDNA under therapy. Mathematical models of HBV infection in CHB under antiviral therapy may need to include infected cell proliferation. Under the assumption that the hepatocyte number is conserved, modeling infected cell proliferation depends on two factors: (1) the probability that an infected cell divides to replace a dead cell and (2) the distribution of cccDNA to its daughter cells. The first factor is influenced by the proportion of infected cells in the liver. If there is a higher proportion of infected cells in the liver, there will be a higher probability that an infected cell (rather than an uninfected cell) divides to compensate for a loss of cells. The second factor refers to whether an infected cell passes on cccDNA to both, one, or none of its daughter cells.

We show how infected cell proliferation can be accounted for in a mathematical model when we assume that infected cells are uniformly distributed through the liver (Eq. (1)). We used the standard mathematical model (Figure 2a) and made two simplifying assumptions: (1) all new infections are prevented by an entry inhibitor such as Myrcludex-B [2] and (2) there is instantaneous proliferation to maintain the hepatocyte number during CHB [37]. Under these assumptions, the mathematical model is given as follows:

$$\frac{dI}{dt} = - \underbrace{\delta I}_{\text{loss via immune mediated cell killing}} - \underbrace{\delta I \frac{I}{H}}_{\text{loss of infected parent cell via proliferation}} + \underbrace{n \cdot \delta I \frac{I}{H}}_{\text{gain of infected daughter cells via cccDNA distribution}} \quad (1)$$

where the probability of picking an infected cell for proliferation is given by the proportion of infected cells in the liver (I/H) and H is the number of hepatocytes in the liver. A gain is given by the cccDNA distribution to n number of daughter cells, where either n can be

Table 3

Estimates of the half-life of HBV-infected cells.

Estimate	Experimental condition	Reference
11–30 d	ADV ^a	[26]
10–100 d	LMV ^a	[25]
9–24 h	LMV ^a	[46]
43 d	LMV/LMV+FCV ^a	[47]
7.2 ± 1.2 d	LMV/LMV+FCV ^a	[32]
7 d	ETV ^a	[45]
7.4 d	CLV ^a	[31]
16.0 ± 1.7 d	IFN/IFN+LMV/LMV PEG-IFN/PEG-IFN+LMV/ADV+LMV ^a	[44]
2.7–75 d	PEG-IFN/PEG-IFN+LMV/LMV ^b	[48]
4.3–56.5 d	PEG-IFN/PEG-IFN+LMV/LMV ^b	[42]
12.1 ± 1.4 d	IFN/IFN+LMV/LMV PEG-IFN/PEG-IFN+LMV/ADV+LMV ^b	[44]

CHB, chronic HBV; HBV, hepatitis B virus; HBsAg, hepatitis B e-antigen. The half-life of HBV-infected cells was estimated in studies of HBV infection in humans. Various therapies were given to patients with CHB, including adefovir (ADV), lamivudine (LMV), famciclovir (FCV), entecavir (ETV), clevudine (CLV), interferon (IFN), and pegylated interferon (PEG-IFN). The treatment groups within each study are denoted by a slash. Estimates are expressed as a range (minimum–maximum), mean ± standard deviation, or mean.

^a HBsAg(+) patients.

^b HBsAg(–) patients.

assumed to be 0, 1, or 2 (none, one, and both, respectively).

This equation and variants of it can be used to predict the time it takes to eradicate cccDNA under various assumptions. Note that these equations can be also applied to model cells with IDNA, although both daughter cells will carry IDNA ($n = 2$).

Mathematical models including immune responses

In CHB, there are anergic and/or exhausted HBV-specific T cells [12,20]. There is a need to better understand the mechanisms involved in HBV-specific T-cell exhaustion and how novel therapies attempt to restore innate and adaptive immune functions [20]. Existing mathematical models of HBV infection with an immune response need to be revisited because they do not capture immune exhaustion and reversal under antiviral therapy. We propose that this can be achieved by developing mathematical models similar to the post-treatment control model for HIV [64].

Conclusion

The success of a therapy for HBV infection depends on the complete elimination of cccDNA, IDNA, HBV DNA, and HBsAg. Any remnant of HBV DNA and cccDNA could lead to viral rebound, which is often the case when treatment is stopped before the infection is entirely eliminated [19,20]. Any remnant of IDNA is also undesirable as it could lead to the generation of HBsAg and hepatocellular carcinoma [9]. Although determining the complete elimination of the virus in vivo may not be possible, theoretical tools such as mathematical modeling may help predict the optimal duration of novel therapies and may be used in the design of ‘response-guided treatment’ of HBV infection.

In this review, we highlighted the usefulness and challenges of mathematical modeling. One main challenge is the high variability in estimates of the half-lives of HBV DNA, HBsAg, infected cells, and cccDNA, which impact the predictions made with the use of mathematical models. Frequent longitudinal patient sampling in combination with improved models may significantly refine estimates of critical parameters and our understanding of HBV infection.

We also proposed future directions in the mathematical modeling of HBV infection, which are largely inspired by progress in modeling of other viral infections. We suggest mathematical models that include intracellular processes of the HBV life cycle to be further developed. This is an approach that has improved the understanding of hepatitis C virus infection [63,65] and the duration of therapy to achieve a cure [66]. Experimental studies of intracellular HBV replication will be crucial for model validation. We also suggest developing

mathematical models with an immune response, as those developed for the post-treatment control of HIV infection [64].

In the future, concerted efforts and collaboration between experimentalists and mathematical modelers may accelerate the progress toward HBV infection cure.

Conflict of interest statement

Nothing declared.

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